

dbt Semantic Layer (MetricFlow)

The problem this solves

Ask three people at a D2C brand "what's our AOV?" and you'll often get three different numbers — because the definition lives inside whatever SQL each dashboard or analyst happens to write. Metrics drift. Nobody's wrong, but nobody agrees. This is the problem a semantic layer fixes: **define each metric once, query it everywhere, get the same answer every time.**

What's here

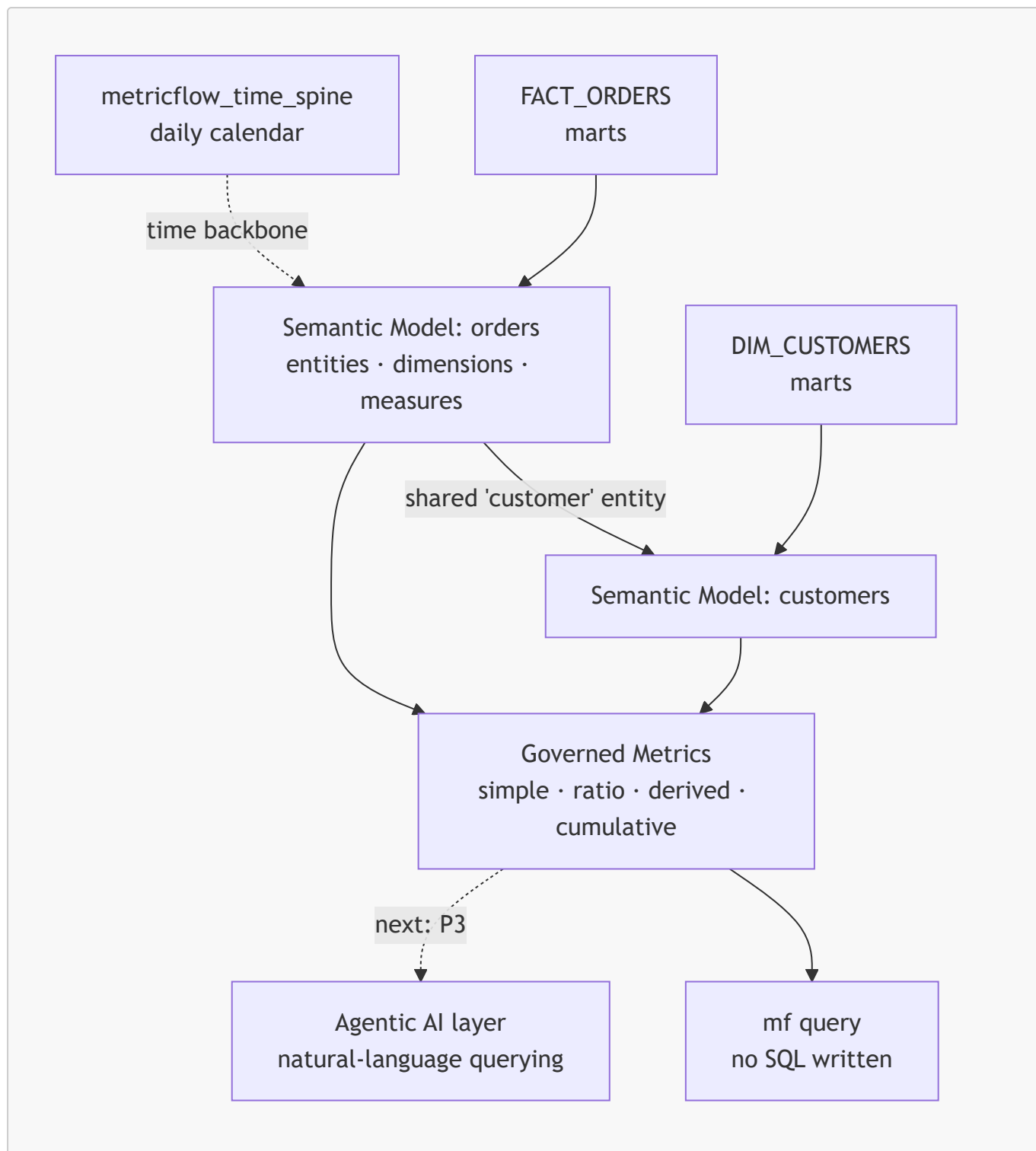
A governed semantic layer over the Petal & Co warehouse, built with dbt + MetricFlow. It sits on top of the existing marts (`FACT_ORDERS`, `DIM_CUSTOMERS`) and exposes business metrics that any tool — a dashboard, an analyst, or (next) an AI agent — can query by name, without writing SQL.

Architecture

Two semantic models, joined automatically on a shared `customer` entity:

Semantic model	Built on	Grain
<code>orders</code>	<code>FACT_ORDERS</code>	one row per order
<code>customers</code>	<code>DIM_CUSTOMERS</code>	one row per customer

A daily **time spine** (`metricflow_time_spine`) backs all time-based and cumulative queries.



Governed metrics

Nine metrics, demonstrating all four MetricFlow types:

Metric	Type	Definition
net_revenue	simple	Sum of net revenue (after discounts + returns)
gross_revenue	simple	Sum of gross revenue
total_orders	simple	Count of orders
total_customers	simple	Count of customers

Metric	Type	Definition
mrr	simple	Net revenue from subscription orders only (subscription-revenue proxy)
aov	ratio	net_revenue / total_orders
return_rate	ratio	returned_orders / total_orders
revenue_per_customer	ratio	net_revenue / total_customers (LTV proxy)
cumulative_net_revenue	cumulative	All-time running total, shown at month-end
revenue_retention_rate	derived	(gross – returned) / gross

How to query (no SQL)

```
# Revenue and AOV by month
mf query --metrics net_revenue,aov --group-by metric_time__month

# Return rate by region
mf query --metrics return_rate --group-by order_record__region

# LTV proxy by region (joins both semantic models automatically)
mf query --metrics revenue_per_customer --group-by customer__region

# All-time running revenue, month-end values, bounded to the data window
mf query --metrics cumulative_net_revenue --group-by metric_time__month \
  --order metric_time__month --start-time 2024-01-01 --end-time 2024-12-31
```

Why it's the foundation for the AI layer

An LLM pointed at raw warehouse tables invents plausible-but-wrong SQL. Pointed at this semantic layer, it can only return metrics that are *correctly defined* — the governed vocabulary is the guardrail that makes natural-language querying trustworthy. This layer is the query interface the agentic AI work builds on.

Build notes / gotchas (for anyone replicating)

- A **time spine** model is required before any time-based metric will parse.
- Entity names must avoid SQL reserved words (**order** → **order_record**).
- Ratio metrics reference **metrics**, not measures (build the simple metric first).
- Cumulative metrics rolled up to a coarser grain take the period's *first* value by default (looks lagged); use **period_agg: last** for a month-end running total.

*Scope note: built on synthetic data. MRR is modeled as net revenue from subscription orders (a proxy; true MRR would use a subscription-grain fact). Churn rate is intentionally deferred to the agentic-AI layer, where predicted churn is queried from **ML_CHURN_RISK_SCORES** rather than mislabeled as an actual-churn metric here.*